Method for Refiguring Pump Performance with Surface Finish

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1. Purpose

The document describes the criteria and the procedure to refigure the hydraulic performances of a centrifugal pump with the surface finish of the hydraulic parts.

The spreadsheet on Mod.465 calculates the rerate of the hydraulic performances of a pump, mainly in terms of head and efficiency, by changing the roughness of the wetted surfaces.

2. Scope.

This procedure is applicable to all the centrifugal pumps.

3. Abbreviations

N	*[rpm]*	*pump rotor speed*
Qbep	*[m3/h or GPM]*	*pump flow rate at best efficiency point and maximum diameter*
Hbep	*[m or ft]*	*pump head per stage at best efficiency point and maximum diameter*
Eta	*[%]*	*pump efficiency*
Etah	*[%]*	*hydraulic efficieny*
POWER	*[kW or HP]*	*pump shaft power*
Nq or Ns	*[/]*	*specific speed of the pump*
Nst	*[/]*	*number of stages*
Nsuct	*[/]*	*number of impeller suction sides*
Ra	*[m or min]*	*arithmetic average roughness of the surface*
Zimp/Hst	*[/]*	*impeller loss contribute as percentage of Head per stage*
Zcas/Hst	*[/]*	*casing loss contribute as percentage of Head per stage*

4. General procedure: description.

The procedure furnishes an estimation of the pump performances refiguring from the evaluation of the losses on the wetted parts due to the change of the surface roughness.

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The procedure is based on the calculation of the friction losses from statistical data, described on Gulich “Centrifugal Pumps “ [1].

The curve in Figure 1 shows the statistical correlation between friction losses contribute on casing and impeller, as function of specific speed of the pump, with different roughness classes, defined on Table 1.

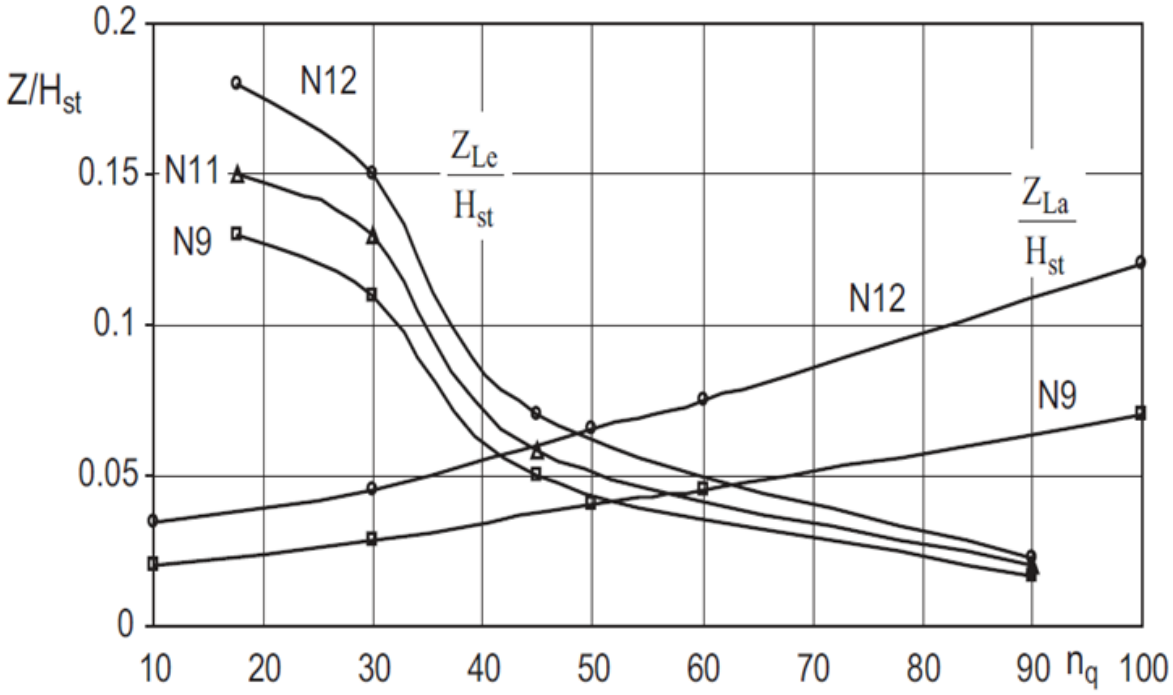

Figure 1. Friction losses contribute on impeller and casing for different roughness classes.

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Table 1. roughness class definition

Roughness class	Ra [micron]	Ra [microin]
N5	0.4	15.7
N6	0.8	31.5
N7	1.6	63.0
N8	3.2	126.0
N9	6.3	248.0
N10	12.5	492.1
N11	25.0	984.3
N12	50.0	1968.5
N13	100.0	3937.0

The percentage Loss/Hst on the graph is correlated with the pump specific speed for static and rotating wetted surfaces, at different values of roughness class.

Two interpolating curves have been calculated as function of specific speed Nq and roughness Ra, one for the impeller and the other for the casing. Each curve in the graph was replicated using polynomial lines, which formulation is described below:

Z/Hst=C0(Nq)+C1(Nq)Ra+C2(Nq)Ra^2

Where C0 , C1 and C2 are coefficient for the polynomial line

Each coefficient is calculated from a polynomial line using another set of parameters, as described below:

C0=A’+B’*Nq+C’*Nq^2+D’*Nq^3*E’*Nq^4*F’*Nq^5*G’*Nq^6
C1=A’’+B’’*Nq+C’’*Nq^2+D’’*Nq^3*E’’*Nq^4*F’’*Nq^5*G’’*Nq^6
C2=A’’’+B’’’*Nq+C’’’*Nq^2+D’’’*Nq^3*E’’’*Nq^4*F’’’*Nq^5*G’’’*Nq^6

the values of the parameters are shown in the matrix in Table 2

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Table 2. matrix of coefficient for the calculation of the losses due to the roughness

casing	A	B	C	D	E	F	G
coeff	Nq^0	Nq^1	Nq^2	Nq^3	Nq^4	Nq^5	Nq^6
C2	-4E-05	8.84E-06	-7.05121E-07	2.49653E-08	-4.35978E-10	3.69E-12	-1.20831E-14
C1	0.002125	-0.00031	2.8935E-05	-1.15511E-06	2.17024E-08	-1.9E-10	6.50315E-13
C0	0.156232	-0.006	0.000608777	-2.97682E-05	6.2479E-07	-5.9E-09	2.07711E-11
imp	A	B	C	D	E	F	G
	Nq^0	Nq^1	Nq^2	Nq^3	Nq^4	Nq^5	Nq^6
C1	0.000335	7.74E-06	-1.13692E-06	5.59926E-08	-1.09924E-09	9.87E-12	-3.33715E-14
C0	0.015424	0.000249	1.32064E-05	-5.30579E-07	1.19145E-08	-1.3E-10	4.88018E-13

Then the calculation of the losses is performed for each component and the total losses contribute of the pump will be predicted. The hydraulic efficiency of the pump is consequently evaluated using the formula below before and after the roughness modification.

Etah=1-(Zimp/Hst+Zcasing/Hst)

The two values of eta, calculated before and after the roughness modification, are used to determine a relative variation of efficiency, using the correlation below:

Delta%=(Etah.old-Etah.new)/Etah.old

The calculated value is used to identity the variation of head per stage at best efficiency point, due to the roughness modification. The used formula is:

DeltaHbep =Hbep*Delta%

Then a coefficient k1 is used to create the correction curve to be applied to the original head, as function of the pump flowrate. It is evaluated through the equation:

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k1=DeltaHbep+Qbep^2

Finally, the new head curve will be determined using the following:

Hnew=H+k1*Q^2

The pump efficiency is calculated using the formula

ETAnew=const*Q*Hnew/Power

5. Description of the calculation tool.

The tool can be used on both SI or US units.

The list of input and output data of the Mod.465 tool are described below.

5.1. Input data.

The input data of the tool are

- the performance curve of the pump (n, Q, H, Power, Efficiency)
- the number of stages Nst and the number of suction sides Nsuct
- the roughness of impeller and casing referred to the input performance data
- the target roughness of impeller and casing

The mask of the input data is shown in Figure 2

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INPUT DATA

unit system			SI
Pump speed			750 RPM
Q	H	P	ETA
m3h	m	kW	%
0.0	215.7	5323.2	0.0
3577.5	214.0	5749.5	36.3
7154.9	209.3	6695.6	60.9
10732.4	198.9	7659.4	75.9
14309.5	182.8	8503.1	83.8
17896.0	162.8	9180.0	86.5
19675.9	151.2	9457.4	85.7
22403.7	129.1	9649.4	81.6

pump name	
numbero of stages	1
number of suction sides	1

roughenss Ra		
	impeller	diffuser/casing
	micron	micron
baseline	12.5	12.5
target	6.3	6.3

Figure 2. Input data.

5.2. Output data

The output data of the tool are

- Specific speed of the pump Nq and Ns
- Performance curve of the pump with target roughness (Q-H-ETA-POWER)

The output data are shown in Figure 3:

OUTPUT DATA

Q	H	P	ETA
m3h	m	kW	%
0.0	215.7	5323.2	0.0
3577.5	214.0	5749.5	36.3
7154.9	209.5	6695.6	61.0
10732.4	199.4	7659.4	76.1
14309.5	183.6	8503.1	84.2
17896.0	164.1	9180.0	87.2
19675.9	152.8	9457.4	86.6
22403.7	131.1	9649.4	82.9

Nq	Ns
36.7	1894

Figure 3. output data tables.

In addition, the tool shows the performances of the pump before and after the refiguring from roughness as shown in Figure 4

The red line refers to the refigured solution, while the black line refers to the original configuration.

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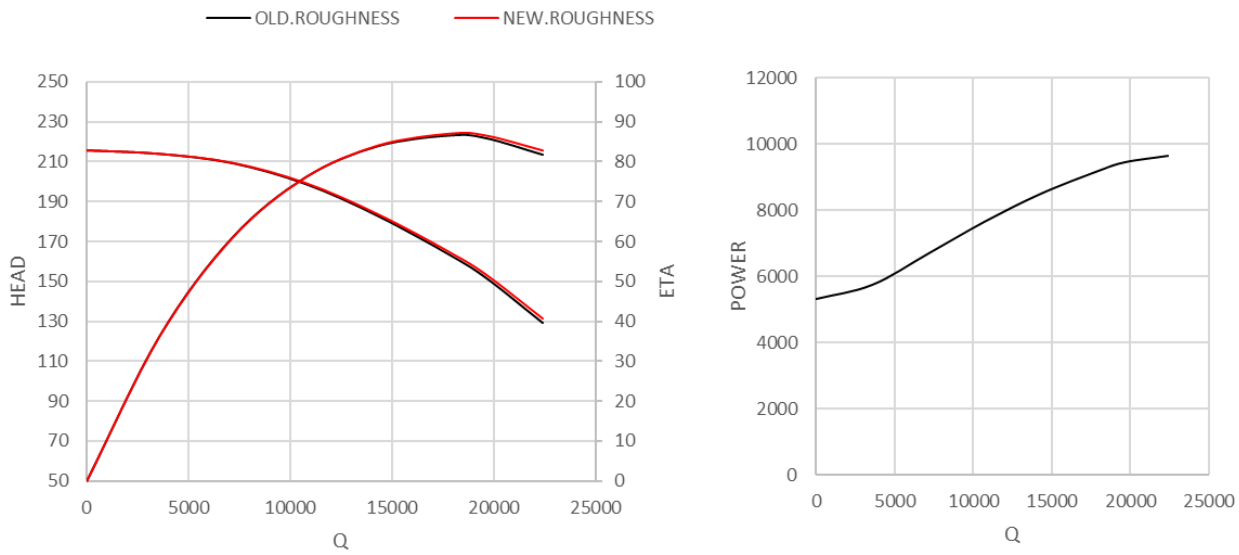

Figure 4. chart of pump performances refiguring from roughness

6. Design recommendations.

The method described in the document is applicable for refiguring the pump performances due to change of roughness in the hydraulic surfaces.

The calculation losses criterion is based on statistical data and it is affected by uncertainness, so it must be used for a high level performance evaluation.

A primary effect of the roughness is to modify the blockage or the hydraulic passages. Typically, when the roughness is high, the blockage due to boundary layer growth effect increase and consequently the internal velocity on the channel rise. This occurs mainly on small sized channels, so, for small pumps, the procedure could underestimate the effective difference on losses between original and new roughness condition. In this case a further investigation with advanced analysis should be processed by R&D team.

7. References

[1]- Gulich "Centrifugal Pumps" ,4th.ed.

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8. Annex

8.1. Case study: BB3 pump type PF80.7 (PROJECT 780810.02)

The pump was initially tested with Ra=22 on the casing and Ra=12.5 on the impeller surfaces.

A second test was performed after polishing the hydraulic surfaces. The final roughness achieved is Ra 6.3 for impeller and Ra 12.5 for casing.

Pump type	BB3
Number of stages	7
rpm	2200
Specific speed Nq	16.5
Reference Test	NC70810.02.01_02-19
Ra imp [micron]	12.5
Ra case [micron]	22
Qbep [m3/h]	94.3
Hbep [m]	399
Eta.test	68.2
Test	NC70810.02.01_03-19
Ra imp [micron]	6.3
Ra case [micron]	12.5
Eta.test	**69**
Eta.calc	**69.3**

The measured pump efficiency was 68.2% during the first test while 69 % is achieved with the polished surfaces (+0.8%). The efficiency calculated using the tool is equal to 69.3%, with a scatter of 0.3%.

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